

PCB DESIGN REQUIREMENTS

Prototype Specification - Rev. 0.15 - Request for Quotation

MECHANICAL ENVELOPE / MOUNTING

The PCB must fit within a maximum available footprint of approximately 50 x 90 mm. A smaller board is preferred where practical. The PCB is expected to be mounted against a substantially flat surface. No specific up/down or rotational operating orientation shall be required. The installation may be vertical, horizontal, inverted, or otherwise oriented as enclosure geometry requires.

The PCB should preferably maintain one substantially flat mounting face. Low-profile components on that face are acceptable where necessary, provided the board can be installed against a flat mounting surface using reasonable standoff spacing. Taller components should be concentrated on the accessible face. All external connectors shall be located on the accessible face.

Four mounting holes, located approximately near the corners, are preferable. Otherwise, two appropriately spaced holes are acceptable. Hole diameter and mounting hardware are not prescribed. No electrical function is intended for the mounting holes.

Installation using vibration-resistant mounting will substantially restrict airflow around the PCB. The design should not depend upon unrestricted convection or a particular PCB orientation for normal thermal management.

REQUIRED

- LiFePO4 cumulative battery-tap measurement through B8
- Minimum 16 single-ended analog acquisition channels
- DS2484 I2C-to-1-Wire interface, or suitable equivalent, with externally accessible 1-Wire DATA and 3.3-V supply
- Minimum four independently controllable logic/control outputs
- Final Output Sense (+) and Final Output Sense (-)
- Native 3.3-V sensor/I2C domain
- Required battery-tap isolation
- 5-V power input
- B0 / Pack negative / System GND shall be the common electrical reference for the PCB electronics, 5-V supply, controller/display, and battery-voltage measurements
- Unused upper battery-tap inputs may remain unconnected
- All ADC channels not assigned to onboard conditioning shall be exposed as general-purpose analog inputs; future inputs will be externally conditioned to meet 3.3-V limits
- Complete PCB design/manufacturing files suitable for public release, modification, and independent manufacture

DESIGNER FREEDOM

Components, ADC topology, isolation implementation, regulators, **non-prescribed** connector grouping/pin assignments, and PCB layout may be changed where required functions are preserved.

We recognize that an experienced PCB designer may identify simpler or better implementations than those used in our prototype, and we are counting on that expertise to achieve robust durability in a harsh, hot, and vibratory environment.

CONNECTORS AND MOUNTING ACCESS

External low-current signal and power connections shall use JST-XH connectors. Any combination of JST-XH connector sizes from 2 through 10 positions may be used as appropriate for the required pinouts and PCB layout. No connector shall exceed 10 positions.

All external connectors shall be top-entry. Edge-mounted or right-angle connectors shall not be used.

The number, sizes, and grouping of connectors are at the designer's discretion, except for the **prescribed** battery-tap and controller/display connectors specified below. Complete connector/pin documentation is required, including the scaling/transfer relationship for every analog input using onboard conditioning.

Where practical, a dedicated 3-position JST-XH connector providing 3.3 V, System GND, and 1-Wire DATA is preferred.

POWER

A regulated 5-V supply is provided through the controller/display interface connector described below. The PCB should generate clean 3.3 V locally for sensor/I2C electronics. Regulator implementation is at the designer's discretion, provided the thermal-management requirements of this specification are met.

HOST CONTROLLER / INTERFACE

The system controller is an Elecrow 2.1-inch Rotary Display, model DHE03921D, incorporating an ESP32-S3. The controller/display remains external to this PCB.

CONTROLLER / DISPLAY INTERFACE AND PCB POWER

Provide a 4-position JST-XH connector carrying 5 V, System GND, SDA, and SCL. Pin order is at the designer's discretion because the mating harness will be wired to suit.

5 V and System GND on this connector are the power input for the PCB and also provide the required power connections for the external controller/display. SDA and SCL provide the native 3.3-V I2C interface to the controller/display.

MINIMUM ANALOG CAPABILITY: 16 SINGLE-ENDED CHANNELS

The working prototype presently uses ADS1115 ADC modules. The ADS1115 architecture is a proven reference implementation.

The final PCB does NOT have to reproduce the existing multi-ADS1115 architecture. If the ADS1115 architecture is retained, four ADS1115 devices provide the required 16-channel capability.

Provide at least four independently controllable logic/control outputs for authorization/control and future expansion. These are separate from the 16 analog measurement channels. ADS1115 ALERT/RDY outputs may fulfill this requirement if that architecture is retained.

A consolidated multi-channel ADC implementation is welcome where it reduces component count, PCB area, cost, or complexity without compromising the stated measurement requirements.

A different ADC or ADC combination may be used if it provides equivalent or superior usable channel count, resolution, accuracy, input compatibility, sampling performance, reliability, and 3.3-V ESP32 interface compatibility.

Battery taps B1 through B8 require dedicated onboard voltage-scaling inputs.

Final Output Sense (+) and Final Output Sense (-) are also dedicated analog measurement inputs. Both nodes must be safely measurable relative to System GND over the full intended operating range. Neither shall be used as a PCB power-supply or ground connection. Their electrical relationship to B0 / System GND may vary with the pack's operating configuration.

These measurements are intended for output-voltage verification, regenerative charging measurements, and comparison with internal pack voltage for future voltage-drop and pack-resistance calculations. The designer may select appropriate input scaling/conditioning compatible with the selected ADC architecture. The established 100k/10k higher-voltage divider topology is an available reference; effective resolution and accuracy should remain suitable for the intended differential measurements. Any alternative scaling or conditioning shall be documented sufficiently to permit implementation of the correct measurement conversion in firmware.

BATTERY TAPS / VOLTAGE DIVIDERS

Provide a dedicated 9-position JST-XH battery-tap connector for an 8S-capable sequential cell-tap arrangement. B0 is a reference connection and is not itself an ADC measurement channel. B1 through B8 are cumulative battery taps requiring onboard voltage scaling.

Pin	Tap	Function	Proven / Intended Divider	Design Maximum
1	B0	Pack negative / System GND	Reference - no ADC channel	-
2	B1	Cumulative tap	47k high / 18k low	3.65 V
3	B2	Cumulative tap	47k high / 18k low	7.30 V
4	B3	Cumulative tap	47k high / 18k low	10.95 V
5	B4	Cumulative tap	100k high / 10k low	14.60 V
6	B5	Cumulative tap	100k high / 10k low	18.25 V
7	B6	Cumulative tap	100k high / 10k low	21.90 V
8	B7	Cumulative tap	100k high / 10k low	25.55 V
9	B8	Highest cumulative tap	100k high / 10k low	29.20 V

The 100k/10k divider topology was established for the existing 24-V measurement and subsequently used for the higher cumulative battery taps in the 3.3-V analog domain. For B6-B8 it is the intended reference topology; the PCB designer shall verify worst-case input voltage, resistor tolerances and voltage ratings, protection, and compatibility with the selected ADC.

Equivalent professional redesign of the divider/protection circuitry is permitted provided the resulting measurement behavior remains compatible with the required ranges and the final scaling/transfer relationship is documented as required elsewhere in this specification.

Where convenient, the battery-tap connector and the 4-position controller/display connector should be located near one another. B0 and System GND are the same electrical node and are intentionally available at both interfaces.

BATTERY-TAP ISOLATION

Source-side isolation of the battery measurement taps is mandatory. When battery-voltage measurement is disabled, the isolation system shall prevent the downstream divider, ADC, protection, and control circuitry from providing a meaningful continuous discharge path from any battery tap. Off-state leakage and long-term cell imbalance during storage must be considered in the design.

The proven prototype uses normally-open mechanical relay contacts to provide physical disconnection of the battery-sense paths during storage. Smaller PCB-mount mechanical relays may be substituted provided equivalent isolation, leakage, voltage-rating, environmental, and low-level signal performance are maintained. Relay coil voltage is at the designer's discretion.

An alternative solid-state implementation may be used only if it provides equivalent storage isolation and suitably negligible off-state leakage. This isolation function shall not be omitted as an optimization.

ENVIRONMENTAL PROTECTION / CONFORMAL COATING

The PCB will operate in a hot, vibratory automotive environment where moisture, contamination, and corrosion resistance are important. The PCB layout and component selection should be compatible with professional conformal coating and should account for contamination-aware spacing and environmental durability.

The designer/manufacture should identify any components or areas requiring masking or exclusion from coating. If conformal coating services are offered without materially extending delivery time, please include the most protective suitable coating option available for this application in the quotation. If coating would require outsourcing or materially extend delivery time, please identify that separately.

PCB IDENTIFICATION / SILKSCREEN

The PCB shall be prominently identified on the silkscreen as "**LiFePO4.**"

The battery-tap connector positions should be clearly identified **B0 through B8.**

DESIGN FILES / PUBLIC RELEASE

The completed PCB design is intended for public release as part of the SNAPACK open-source project under the project's published licensing terms. The public project repository is <https://github.com/secarider/SNAPACK>.

Development has progressed rapidly during the recent prototyping phase, and some documentation in the public repository has not yet been updated to reflect the current hardware configuration. Where repository material differs from this specification, this specification shall be considered current for purposes of PCB design and quotation.

Upon completion, provide the complete PCB design and manufacturing package, including editable schematic and PCB-layout source files, Gerber fabrication files, drill files, BOM, and pick-and-place/centroid data as applicable. The delivered design files must be suitable for public release, future modification, and manufacture by any PCB manufacturer of our choosing.

Any pre-existing proprietary design assets, libraries, or other restrictions that would prevent or limit public release, modification, or independent manufacture of the completed PCB design must be identified before work begins.

The PCB designer/manufacture is welcome to include reasonable company identification, attribution, or promotional marking on the PCB silkscreen, within the design/manufacturing files, and/or in documentation accompanying the publicly released PCB files, provided it does not restrict or interfere with public release, modification, or independent manufacture of the design.

REFERENCE MATERIAL

Current working ESP32-S3 firmware and additional prototype documentation are available upon request. Firmware can be provided to confirm I2C devices, ADC behavior, sensor interfaces, and integration requirements.

PROTOTYPE QUANTITY

Please quote five assembled prototype boards.

If prototype quantity materially affects total cost, please also indicate the cost at lower quantities. If additional boards have relatively low incremental cost after design/setup, please indicate the incremental cost for additional boards.